Roya Akrami

+1 972-795-5871 · roya_akrami@mines.edu · royasakrami7@gmail.com
Golden, Colorado, United States

EDUCATIONAL EXPERIENCE

Colorado School of Mines **Graduating May 2027**
Bachelor of Science, Electrical Engineering

- Conducted hardware-integration research using Verilog, C, and MPLAB X IDE, worked with custom PCBs, FPGA Lite systems, and oscilloscopes for signal analysis and embedded control

KEY ACCOMPLISHMENTS

- **Exospheric Hydrogen Mapping - NASA Internship -** Processed H-Lya data with MATLAB and a 28-layer ResNet; paper in process of development using NASA Roman Telescope data for the Carruthers Geocorona Observatory.
- **American Geophysical Union Conference 2025 -** Presented project, "Development of a Regression Model of Exospheric Lyman-Alpha Emission Using Neural Networks," communicating NASA internship research in space-based machine learning.
- **Vibrational Circular Dichroism (VCD) Spectroscopy Research -** Contributed figures and annotated references to a vibrational circular dichroism (VCD) literature review and the corresponding article, "*Rapid Vibrational Circular Dichroism Spectroscopy via Synchronized Photoelastic Modulator–Quantum Cascade Laser Integration.*"
- **Represented Colorado School of Mines in Washington, D.C. -** As part of the Council on Undergraduate Research advocacy program, met with U.S. Senators to advocate for federal research funding during nationwide funding cuts.

PROFESSIONAL EXPERIENCE

Research Assistant **March 2025 - Present**
NASA Goddard Space Flight Center

- Research paper in process of development on extracting and analyzing H-Lya emissions in Earth's exosphere using data from the Nancy Grace Roman Telescope to advance exospheric mapping methods for the Carruthers Geocorona Observatory.
- Processed H-Lya intensity data in JupyterHub and MATLAB through a 28-layer residual neural network to model and predict future emission patterns.

Research Assistant **Oct 2023 - Oct 2025**
Phal Lab, Colorado School of Mines

- Designed figures, wrote annotations, and gathered references for paper discussing vibrational circular dichroism detection of Alzheimer's biomarkers, resulting in a paper in process of publication and a published article.
- Updated design and content of lab website using SCSS, HTML, and Ruby languages to heighten its quality.
- Designed and implemented a 2D LiDAR-based mapping system on a LattePanda computer using Ubuntu and ROS RViz for real-time visualization and spatial mapping.

Undergraduate Research Ambassador **Aug 2024 - Present**
Colorado School of Mines

- Selected to represent Colorado School of Mines as a student ambassador for undergraduate research in Washington, D.C.; funded by the STR Program to meet with U.S. senators and discuss research funding and policy.
- Presented to students on the value of undergraduate research and provided mentorship through email outreach, and authored articles highlighting student research initiatives.

Society of Women in Research **Feb 2025 - Present**
Founder and President

- Founded and led the only organization at Colorado School of Mines dedicated to closing the gender gap in research participation.
- Organized weekly meetings, cold-email workshops, and scholarship information sessions to connect female students with research opportunities. Presented organization to 100 students and parents.

LANGUAGES

C2-level French • C++ • Python • MATLAB • C+ JupyterHub • MPLAB x IDE • Inkscape • Excel • LaTeX • Verilog

HONORS & AWARDS

ThinkSwiss Scholarship **February 2026**

- Awarded $7,000 to conduct research overseas at ETH Zurich of Switzerland.

Travel Grant from Colorado School of Mines **December 2025**

- Awarded $400 for travel expenses to attend American Geophysical Union conference to present exospheric machine learning research conducted at the NASA Goddard Space Flight Center.

NASA and Universities Space Research Association **Spring 2025**

- Awarded a ~$10,000 undergraduate research stipend to conduct NASA-funded research.

Students Transforming Through Research Conference **Spring 2025**

- Funded to attend a research policymaking conference in Washington D.C to represent Mines as an Undergraduate Research Ambassador.

Summer Undergraduate Research Fellowship **June - Aug 2024**

- $4000 stipend to conduct faculty-mentored summer research, granted by Colorado School of Mines.

3rd Place at Spring Undergraduate Research Symposium **April 2024**

- Presented 12-minute presentation as a first-year student of vibrational circular dichroism designs at Mines Research Symposium, won 3rd place out of all presenters.

First-Year Innovation & Research Scholar Training **Aug 2023 - May 2024**

- $1000 stipend to work in a research lab, granted by Colorado School of Mines.

The Provost Award **Aug 2023 - May 2027**

- $10,000-per-year scholarship granted by Colorado School of Mines